JOINT ENTRANCE EXAMINATION (MAIN) JULY -2022
(CANDIDATE RESPONSE SHEET)

Paper/Subject	B.E/B.Tech.(Paper 1)
Exam Date	26 Jul 2022
Exam Slot	2

Question ID:144961
Topic Name: Mathematics-Section A
Question: The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is:

A 4

B 5

C 6

D 8

Answer Given By Candidate:**C**

Question ID:144962
Topic Name: Mathematics-Section A
Question: If $z = x + iy$ satisfies $|z| - 2 = 0$ and $|z - i| - |z + 5i| = 0$, then

A $x + 2y - 4 = 0$

B $x^2 + y - 4 = 0$

C $x + 2y + 4 = 0$

D $x^2 - y + 3 = 0$

Answer Given By Candidate:**C**

Question ID:144963
Topic Name: Mathematics-Section A
Question: Let $A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $B = \begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix}$, then the value of $A'BA$ is:

A 1224

B 1042

C 540

D 539

Answer Given By Candidate:**Not Attempted**

Question ID:144964
Topic Name: Mathematics-Section A

Question: $\sum_{\substack{i,j=0 \\ i \neq j}}^{n} {}^{n}C_i \, {}^{n}C_j$ is equal to

A $2^{2n} - {}^{2n}C_n$

B $2^{2n-1} - {}^{2n-1}C_{n-1}$

C $2^{2n} - \frac{1}{2}\,{}^{2n}C_n$

D $2^{n-1} + {}^{2n-1}C_n$

Answer Given By Candidate:**Not Attempted**

Question ID:144965
Topic Name: Mathematics-Section A

Question: Let P and Q be any points on the curves $(x-1)^2 + (y+1)^2 = 1$ and $y = x^2$, respectively. The distance between P and Q is minimum for some value of the abscissa of P in the interval

A $\left(0, \frac{1}{4}\right)$

B $\left(\frac{1}{2}, \frac{3}{4}\right)$

C $\left(\frac{1}{4}, \frac{1}{2}\right)$

D $\left(\frac{3}{4}, 1\right)$

Answer Given By Candidate:**Not Attempted**

Question ID:144966
Topic Name: Mathematics-Section A

Question: If the maximum value of a, for which the function $f_a(x) = \tan^{-1} 2x - 3ax + 7$ is non-decreasing in $\left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$, is $\overline{a}$, then $f_{\overline{a}}\left(\frac{\pi}{8}\right)$ is equal to

A $8 - \frac{9\pi}{4(9+\pi^2)}$

B $8 - \frac{4\pi}{9(4+\pi^2)}$

C $8\left(\frac{1+\pi^2}{9+\pi^2}\right)$

D $8 - \frac{\pi}{4}$

Answer Given By Candidate: **Not Attempted**

Question ID: 144967
Topic Name: Mathematics-Section A

Question: Let $\beta = \lim\limits_{x \to 0} \frac{\alpha x - (e^{3x} - 1)}{\alpha x (e^{3x} - 1)}$ for some $\alpha \in \mathbb{R}$. Then the value of $\alpha + \beta$ is :

A $\frac{14}{5}$

B $\frac{3}{2}$

C $\frac{5}{2}$

D $\frac{7}{2}$

Answer Given By Candidate: **Not Attempted**

Question ID: 144968
Topic Name: Mathematics-Section A

Question: The value of $\log_e 2 \frac{d}{dx}\left(\log_{\cos x} \operatorname{cosec} x\right)$ at $x = \frac{\pi}{4}$ is

A $-2\sqrt{2}$

B $2\sqrt{2}$

C -4

D 4

Answer Given By Candidate:**A**

Question ID:144969
Topic Name: Mathematics-Section A
Question: $\int_0^{20\pi} (|\sin x| + |\cos x|)^2 \, dx$ is equal to

A $10(\pi + 4)$

B $10(\pi + 2)$

C $20(\pi - 2)$

D $20(\pi + 2)$

Answer Given By Candidate:**D**

Question ID:1449610
Topic Name: Mathematics-Section A
Let the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1 - x^2}}$, $x \in (-1, 1)$ pass through the origin. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$ is

Question: equal to

A $\frac{\pi}{3} - \frac{1}{4}$

B $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$

C $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$

D $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Answer Given By Candidate:**Not Attempted**

Question ID:1449611
Topic Name: Mathematics-Section A

Question: The acute angle between the pair of tangents drawn to the ellipse $2x^2 + 3y^2 = 5$ from the point (1, 3) is

A $\tan^{-1}\left(\frac{16}{7\sqrt{5}}\right)$

B $\tan^{-1}\left(\frac{24}{7\sqrt{5}}\right)$

C $\tan^{-1}\left(\frac{32}{7\sqrt{5}}\right)$

D $\tan^{-1}\left(\frac{3+8\sqrt{5}}{35}\right)$

Answer Given By Candidate: **B**

Question ID: 1449612
Topic Name: Mathematics-Section A

Question: The equation of a common tangent to the parabolas $y = x^2$ and $y = -(x-2)^2$ is

A $y = 4(x-2)$

B $y = 4(x-1)$

C $y = 4(x+1)$

D $y = 4(x+2)$

Answer Given By Candidate: **B**

Question ID: 1449613
Topic Name: Mathematics-Section A

Question: Let the abscissae of the two points P and Q on a circle be the roots of $x^2 - 4x - 6 = 0$ and the ordinates of P and Q be the roots of $y^2 + 2y - 7 = 0$. If PQ is a diameter of the circle $x^2 + y^2 + 2ax + 2by + c = 0$, then the value of $(a + b - c)$ is _____.

A 12

B 13

C 14

D 16

Answer Given By Candidate: **A**

Question ID:1449614
Topic Name: Mathematics-Section A

Question: If the line $x - 1 = 0$ is a directrix of the hyperbola $kx^2 - y^2 = 6$, then the hyperbola passes through the point

A $\left(-2\sqrt{5}, 6\right)$

B $\left(-\sqrt{5}, 3\right)$

C $\left(\sqrt{5}, -2\right)$

D $\left(2\sqrt{5}, 3\sqrt{6}\right)$

Answer Given By Candidate:**C**

Question ID:1449615
Topic Name: Mathematics-Section A

Question: A vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}, \hat{i}+\hat{j}$ and the plane determined by the vectors $\hat{i}-\hat{j}, \hat{i}+\hat{k}$. The obtuse angle between $\vec{a}$ and the vector $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$ is

A $\frac{3\pi}{4}$

B $\frac{2\pi}{3}$

C $\frac{4\pi}{5}$

D $\frac{5\pi}{6}$

Answer Given By Candidate:**Not Attempted**

Question ID:1449616
Topic Name: Mathematics-Section A

Question: If $0 < x < \frac{1}{\sqrt{2}}$ and $\frac{\sin^{-1} x}{\alpha} = \frac{\cos^{-1} x}{\beta}$, then a value of $\sin\left(\frac{2\pi\alpha}{\alpha+\beta}\right)$ is

A $4\sqrt{\left(1-x^2\right)}\left(1-2x^2\right)$

B $4x\sqrt{\left(1-x^2\right)}\left(1-2x^2\right)$

C $2x\sqrt{\left(1-x^2\right)}\left(1-4x^2\right)$

D $4\sqrt{\left(1-x^2\right)}\left(1-4x^2\right)$

Answer Given By Candidate:**B**

Question ID:1449617
Topic Name: Mathematics-Section A
Question: Negation of the Boolean expression $p \Leftrightarrow (q \Rightarrow p)$ is

A $(\sim p) \wedge q$

B $p \wedge (\sim q)$

C $(\sim p) \vee (\sim q)$

D $(\sim p) \wedge (\sim q)$

Answer Given By Candidate:**D**

Question ID:1449618
Topic Name: Mathematics-Section A

Let X be a binomially distributed random variable with mean 4 and variance $\frac{4}{3}$.

Question: Then, $54\,P(X \leq 2)$ is equal to

A $\frac{73}{27}$

B $\frac{146}{27}$

C $\frac{146}{81}$

D $\frac{126}{81}$

Answer Given By Candidate:**Not Attempted**

Question ID:1449619
Topic Name: Mathematics-Section A

Question: The integral $\int \frac{\left(1-\frac{1}{\sqrt{3}}\right)(\cos x-\sin x)}{\left(1+\frac{2}{\sqrt{3}}\sin 2x\right)}dx$ is equal to

A $\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}\right|+C$

B $\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{3}\right)}\right|+C$

C $\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}\right|+C$

D $\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}-\frac{\pi}{6}\right)}\right|+C$

Answer Given By Candidate: **Not Attempted**

Question ID: 1449620
Topic Name: Mathematics-Section A
Question: The area bounded by the curves $y = |x^2 - 1|$ and $y = 1$ is

A $\frac{2}{3}\left(\sqrt{2}+1\right)$

B $\frac{4}{3}\left(\sqrt{2}-1\right)$

C $2\left(\sqrt{2}-1\right)$

D $\frac{8}{3}\left(\sqrt{2}-1\right)$

Answer Given By Candidate:**D**

Question ID:1449621
Topic Name: Mathematics-Section B
Question: Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{3, 6, 7, 9\}$. Then the number of elements in the set $\{C \subseteq A : C \cap B \neq \phi\}$ is _____

Answer Given By Candidate:**Not Attempted**

Question ID:1449622
Topic Name: Mathematics-Section B
Question: The largest value of a, for which the perpendicular distance of the plane containing the lines $\vec{r} = \left(\hat{i}+\hat{j}\right)+\lambda\left(\hat{i}+a\hat{j}-\hat{k}\right)$ and $\vec{r} = \left(\hat{i}+\hat{j}\right)+\mu\left(-\hat{i}+\hat{j}-a\hat{k}\right)$ from the point (2, 1, 4) is $\sqrt{3}$, is_____.

Answer Given By Candidate:**Not Attempted**

Question ID:1449623
Topic Name: Mathematics-Section B
Question: Numbers are to be formed between 1000 and 3000, which are divisible by 4, using the digits 1, 2, 3, 4, 5 and 6 without repetition of digits. Then the total number of such numbers is _____.

Answer Given By Candidate:**30**

Question ID:1449624
Topic Name: Mathematics-Section B
Question: If $\sum_{k=1}^{10} \frac{k}{k^4+k^2+1} = \frac{m}{n}$, where m and n are co-prime, then $m + n$ is equal to _____

Answer Given By Candidate:**Not Attempted**

Question ID:1449625
Topic Name: Mathematics-Section B
Question: If the sum of solutions of the system of equations $2\sin^2\theta - \cos2\theta = 0$ and $2\cos^2\theta + 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is $k\pi$, then k is equal to _____.

Answer Given By Candidate:**3**

Question ID:1449626
Topic Name: Mathematics-Section B

Question: The mean and standard deviation of 40 observations are 30 and 5 respectively. It was noticed that two of these observations 12 and 10 were wrongly recorded. If σ is the standard deviation of the data after omitting the two wrong observations from the data, then $38\sigma^2$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1449627
Topic Name: Mathematics-Section B

Question: The plane passing through the line L: $lx - y + 3(1 - l)z = 1$, $x + 2y - z = 2$ and perpendicular to the plane $3x + 2y + z = 6$ is $3x - 8y + 7z = 4$. If θ is the acute angle between the line L and the y-axis, then $415\cos^2\theta$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1449628
Topic Name: Mathematics-Section B

Question: Suppose $y = y(x)$ be the solution curve to the differential equation $\frac{dy}{dx} - y = 2 - e^{-x}$ such that $\lim\limits_{x\to\infty} y(x)$ is finite. If a and b are respectively the x - and y - intercepts of the tangent to the curve at $x = 0$, then the value of $a - 4b$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1449629
Topic Name: Mathematics-Section B

Question: Different A.P.'s are constructed with the first term 100, the last term 199, and integral common differences. The sum of the common differences of all such A.P.'s having at least 3 terms and at most 33 terms is _____.

Answer Given By Candidate:**42**

Question ID:1449630
Topic Name: Mathematics-Section B

Question: The number of matrices $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $a, b, c, d \in \{-1, 0, 1, 2, 3, \ldots\ldots, 10\}$, such that $A = A^{-1}$, is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1449631
Topic Name: Physics-Section A

Question: Two projectiles are thrown with same initial velocity making an angle of 45° and 30° with the horizontal respectively. The ratio of their respective ranges will be :

A $1:\sqrt{2}$

B $\sqrt{2}:1$

C $2:\sqrt{3}$

D $\sqrt{3}:2$

Answer Given By Candidate:**C**

Question ID:1449632
Topic Name: Physics-Section A

Question: In a Vernier Calipers, 10 divisions of Vernier scale is equal to the 9 divisions of main scale. When both jaws of Vernier calipers touch each other, the zero of the Vernier scale is shifted to the left of zero of the main scale and 4th Vernier scale division exactly coincides with the main scale reading. One main scale division is equal to 1 mm. While measuring diameter of a spherical body, the body is held between two jaws. It is now observed that zero of the Vernier scale lies between 30 and 31 divisions of main scale reading and 6th Vernier scale division exactly coincides with the main scale reading. The diameter of the spherical body will be :

A 3.02 cm

B 3.06 cm

C 3.10 cm

D 3.20 cm

Answer Given By Candidate:**Not Attempted**

Question ID:1449633
Topic Name: Physics-Section A

Question: A ball of mass 0.15 kg hits the wall with its initial speed of 12 ms^{-1} and bounces back without changing its initial speed. If the force applied by the wall on the ball during the contact is 100 N, calculate the time duration of the contact of ball with the wall.

A 0.018 s

B 0.036 s

C 0.009 s

D 0.072 s

Answer Given By Candidate:**B**

Question ID:1449634
Topic Name: Physics-Section A

Question: A body of mass 8 kg and another of mass 2 kg are moving with equal kinetic energy. The ratio of their respective momenta will be :

A 1:1

B 2:1

C 1:4

D 4:1

Answer Given By Candidate: **B**

Question ID: 1449635
Topic Name: Physics-Section A

Question: Two uniformly charged spherical conductors A and B of radii 5 mm and 10 mm are separated by a distance of 2 cm. If the spheres are connected by a conducting wire, then in equilibrium condition, the ratio of the magnitudes of the electric fields at the surface of the sphere A and B will be :

A 1:2

B 2:1

C 1:1

D 1:4

Answer Given By Candidate: **Not Attempted**

Question ID: 1449636
Topic Name: Physics-Section A

Question: The oscillating magnetic field in a plane electromagnetic wave is given by $B_y = 5 \times 10^{-6} \sin 1000\pi(5x - 4 \times 10^8 t)T$. The amplitude of electric field will be :

A $15 \times 10^2\ Vm^{-1}$

B $5 \times 10^{-6}\ Vm^{-1}$

C $16 \times 10^{12}\ Vm^{-1}$

D $4 \times 10^2\ Vm^{-1}$

Answer Given By Candidate: **A**

Question ID: 1449637
Topic Name: Physics-Section A

Question: Light travels in two media M_1 and M_2 with speeds $1.5 \times 10^8\ ms^{-1}$ and $2.0 \times 10^8\ ms^{-1}$ respectively. The critical angle between them is :

A $\tan^{-1}\left(\frac{3}{\sqrt{7}}\right)$

B $\tan^{-1}\left(\frac{2}{3}\right)$

C $\cos^{-1}\left(\frac{3}{4}\right)$

D $\sin^{-1}\left(\frac{2}{3}\right)$

Answer Given By Candidate:**A**

Question ID:1449638
Topic Name: Physics-Section A

Question: A body is projected vertically upwards from the surface of earth with a velocity equal to one third of escape velocity. The maximum height attained by the body will be :
(Take radius of earth = 6400 km and g=10 ms^{-2})

A 800 km

B 1600 km

C 2133 km

D 4800 km

Answer Given By Candidate:**Not Attempted**

Question ID:1449639
Topic Name: Physics-Section A

Question: The maximum and minimum voltage of an amplitude modulated signal are 60 V and 20 V respectively. The percentage modulation index will be :

A 0.5%

B 50%

C 2%

D 30%

Answer Given By Candidate:**B**

Question ID:1449640
Topic Name: Physics-Section A

Question: A nucleus of mass M at rest splits into two parts having masses $\frac{M'}{3}$ and $\frac{2M'}{3}$ $(M' < M)$. The ratio of de Broglie wavelength of two parts will be :

A 1:2

B 2:1

C 1:1

D 2:3

Answer Given By Candidate: **C**

Question ID: 1449641
Topic Name: Physics-Section A

Question: An ice cube of dimensions 60 cm × 50 cm × 20 cm is placed in an insulation box of wall thickness 1 cm. The box keeping the ice cube at 0°C of temperature is brought to a room of temperature 40°C. The rate of melting of ice is approximately :
(Latent heat of fusion of ice is 3.4×10^5 J kg^{-1} and thermal conducting of insulation wall is 0.05 $Wm^{-1}°C^{-1}$)

A 61×10^{-3} kg s^{-1}

B 61×10^{-5} kg s^{-1}

C 208 kg s^{-1}

D 30×10^{-5} kg s^{-1}

Answer Given By Candidate: **Not Attempted**

Question ID: 1449642
Topic Name: Physics-Section A

Question: A gas has n degrees of freedom. The ratio of specific heat of gas at constant volume to the specific heat of gas at constant pressure will be :

A $\frac{n}{n+2}$

B $\frac{n+2}{n}$

C $\frac{n}{2n+2}$

D $\frac{n}{n-2}$

Answer Given By Candidate:**B**

Question ID:1449643
Topic Name: Physics-Section A

Question: A transverse wave is represented by $y = 2\sin(\omega t - kx)$ cm. The value of wavelength (in cm) for which the wave velocity becomes equal to the maximum particle velocity, will be :

A 4π

B 2π

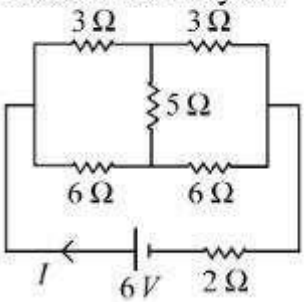
C π

D 2

Answer Given By Candidate:**A**

Question ID:1449644
Topic Name: Physics-Section A

Question: A battery of $6V$ is connected to the circuit as shown below. The current I drawn from the battery is :



A $1A$

B $2A$

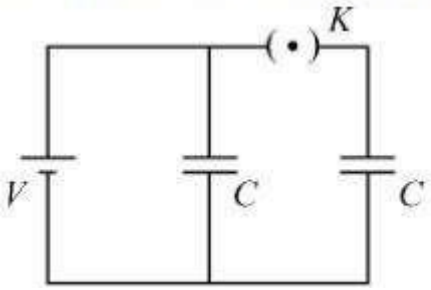
C $\frac{6}{11}A$

D $\frac{4}{3}A$

Answer Given By Candidate:**A**

Question ID:1449645
Topic Name: Physics-Section A

A source of potential difference V is connected to the combination of two identical capacitors as shown in the figure. When key 'K' is closed, the total energy stored across the combination is E_1. Now key 'K' is opened and dielectric of dielectric constant 5 is introduced between the plates of the capacitors. The total energy stored across the combination is now E_2. The ratio E_1/E_2 will be :



Question:

A $\frac{1}{10}$

B $\frac{2}{5}$

C $\frac{5}{13}$

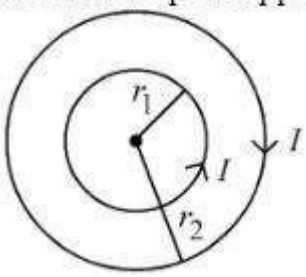
D $\frac{5}{26}$

Answer Given By Candidate: **B**

Question ID: 1449646

Topic Name: Physics-Section A

Two concentric circular loops of radii r_1=30 cm and r_2=50 cm are placed in X-Y plane as shown in the figure. A current $I = 7A$ is flowing through them in the direction as shown in figure. The net magnetic moment of this system of two circular loops is approximately :



Question:

A $\frac{7}{2}\hat{k}\,Am^2$

B $-\frac{7}{2}\hat{k}\,Am^2$

C $7\hat{k}\,Am^2$

D

$-7\,\hat{k}\,Am^2$

Answer Given By Candidate:**A**

Question ID:1449647
Topic Name: Physics-Section A

A velocity selector consists of electric field $\vec{E} = E\hat{k}$ and magnetic field $\vec{B} = B\hat{j}$ with $B = 12\ mT$. The value of E required for an electron of energy $728\ eV$ moving along the positive x-axis to pass undeflected is :

Question: (Given, mass of electron $= 9.1 \times 10^{-31}$kg)

A $192\ \text{kVm}^{-1}$

B $192\ \text{mVm}^{-1}$

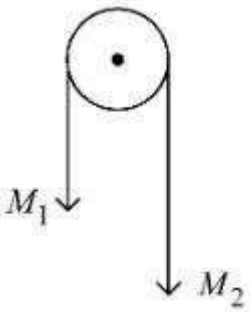
C $9600\ \text{kVm}^{-1}$

D $16\ \text{kVm}^{-1}$

Answer Given By Candidate:**A**

Question ID:1449648
Topic Name: Physics-Section A

Two masses M_1 and M_2 are tied together at the two ends of a light inextensible string that passes over a frictionless pulley. When the mass M_2 is twice that of M_1, the acceleration of the system is a_1. When the mass M_2 is thrice that of M_1, the acceleration of the system is a_2. The ratio $\frac{a_1}{a_2}$ will be :



Question:

A $\frac{1}{3}$

B $\frac{2}{3}$

C $\frac{3}{2}$

D

$\frac{1}{2}$

Answer Given By Candidate:**B**

Question ID:1449649
Topic Name: Physics-Section A

Question: Mass numbers of two nuclei are in the ratio of 4:3. Their nuclear densities will be in the ratio of

A 4 : 3

B $\left(\frac{3}{4}\right)^{\frac{1}{3}}$

C 1 : 1

D $\left(\frac{4}{3}\right)^{\frac{1}{3}}$

Answer Given By Candidate:**C**

Question ID:1449650
Topic Name: Physics-Section A

Question: The area of cross section of the rope used to lift a load by a crane is $2.5 \times 10^{-4} m^2$. The maximum lifting capacity of the crane is 10 metric tons. To increase the lifting capacity of the crane to 25 metric tons, the required area of cross section of the rope should be : (take g=10 ms^{-2})

A $6.25 \times 10^{-4} \mathrm{m}^2$

B $10 \times 10^{-4} \mathrm{m}^2$

C $1 \times 10^{-4} \mathrm{m}^2$

D $1.67 \times 10^{-4} \mathrm{m}^2$

Answer Given By Candidate:**A**

Question ID:1449651
Topic Name: Physics-Section B

Question: If $\vec{A} = \left(2\hat{i}+3\hat{j}-\hat{k}\right)$ m and $\vec{B} = \left(\hat{i}+2\hat{j}+2\hat{k}\right)$ m . The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be _____ m.

Answer Given By Candidate:**Not Attempted**

Question ID:1449652
Topic Name: Physics-Section B

Question: The radius of gyration of a cylindrical rod about an axis of rotation perpendicular to its length and passing through the center will be _____ m.

Given, the length of the rod is $10\sqrt{3}$ m.

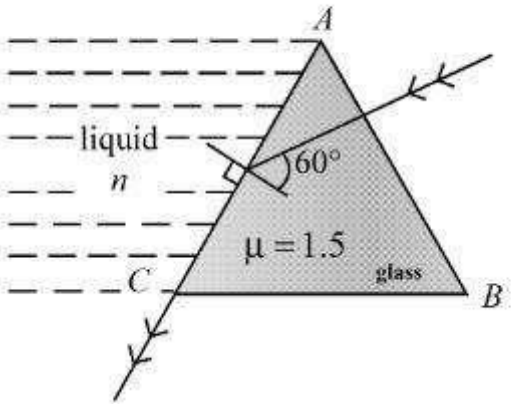
Answer Given By Candidate:**5**

Question ID:1449653
Topic Name: Physics-Section B

Question: In the given figure, the face AC of the equilateral prism is immersed in a liquid of refractive index 'n'. For incident angle 60° at the side AC, the refracteel light beam just grazes along face AC. The refractive index of the liquid $n = \frac{\sqrt{x}}{4}$.

The value of x is _____.

(Given refractive index of glass = 1.5)



Answer Given By Candidate:**27**

Question ID:1449654
Topic Name: Physics-Section B

Question: Two lighter nuclei combine to form a comparatively heavier nucleus by the relation given below :

$${}_1^2X + {}_1^2X = {}_2^4Y$$

The binding energies per nucleon for ${}_1^2X$ and ${}_2^4Y$ are 1.1 MeV and 7.6 MeV respectively. The energy released in this process is _____ MeV.

Answer Given By Candidate:**Not Attempted**

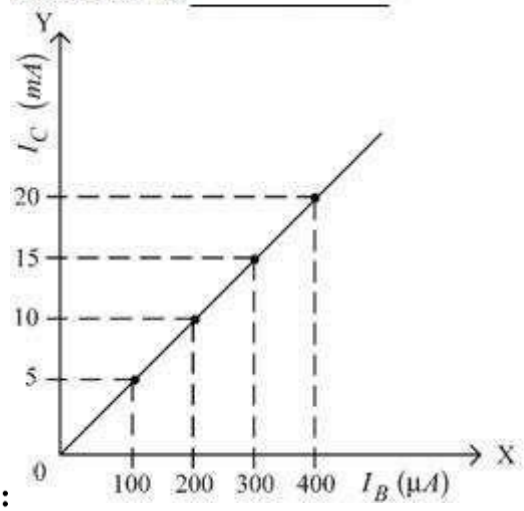
Question ID:1449655
Topic Name: Physics-Section B

Question: A uniform heavy rod of mass 20 kg, cross sectional area 0.4 m^2 and length 20 m is hanging from a fixed support. Neglecting the lateral contraction, the elongation in the rod due to its own weight is $x \times 10^{-9}$ m. The value of x is _____.
(Given, young modulus $Y=2\times10^{11}\ Nm^{-2}$ and $g=10\ ms^{-2}$)

Answer Given By Candidate: **50**

Question ID: 1449656
Topic Name: Physics-Section B

Question: The typical transfer characteristics of a transistor in CE configuration is shown in figure. A load resistor of 2 $k\Omega$ is connected in the collector branch of the circuit used. The input resistance of the transistor is 0.50 $k\Omega$. The voltage gain of the transistor is _____.



Answer Given By Candidate: **100**

Question ID: 1449657
Topic Name: Physics-Section B

Question: Three point charges of magnitude 5μC, 0.16μC and 0.3μC are located at the vertices A, B, C of a right angled triangle whose sides are AB = 3 cm, BC= $3\sqrt{2}$ cm and CA = 3 cm and point A is the right angle corner. Charge at point A experiences _____ N of electrostatic force due to the other two charges.

Answer Given By Candidate: **Not Attempted**

Question ID: 1449658
Topic Name: Physics-Section B

Question: In a coil of resistance 8 Ω,the magnetic flux due to an external magnetic field varies with time as $\phi = \frac{2}{3}\left(9-t^2\right)$. The value of total heat produced in the coil, till the flux becomes zero, will be _____ J.

Answer Given By Candidate: **Not Attempted**

Question ID:1449659
Topic Name: Physics-Section B

Question: A potentiometer wire of length 300 cm is connected in series with a resistance 780 Ω and a standard cell of emf 4V. A constant current flows through potentiometer wire. The length of the null point for cell of emf 20 mV is found to be 60 cm. The resistance of the potentiometer wire is _____Ω.

Answer Given By Candidate:**Not Attempted**

Question ID:1449660
Topic Name: Physics-Section B

Question: As per given figures, two springs of spring constants k and $2k$ are connected to mass m. If the period of oscillation in figure (a) is 3s, then the period of oscillation in figure (b) will be $\sqrt{x}\,s$. The value of x is _____.

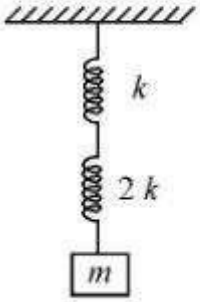


figure (a)

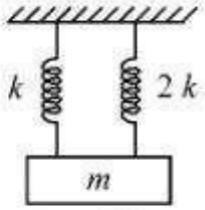


figure (b)

Answer Given By Candidate:**3**

Question ID:1449661
Topic Name: Chemistry-Section A

Question: Hemoglobin contains 0.34% of iron by mass. The number of Fe atoms in 3.3 g of hemoglobin is
(Given: Atomic mass of Fe is 56 u, $N_A = 6.022 \times 10^{23}\ mol^{-1}$.)

A 1.21×10^{5}

B 12.0×10^{16}

C 1.21×10^{20}

D 3.4×10^{22}

Answer Given By Candidate:**C**

Question ID:1449662
Topic Name: Chemistry-Section A

Arrange the following in increasing order of their covalent character.

A. CaF_2
B. $CaCl_2$
C. $CaBr_2$
D. CaI_2

Question: Choose the correct answer from the options given below.

A B < A < C < D

B A < B < C < D

C A < B < D < C

D A < C < B < D

Answer Given By Candidate: **B**

Question ID: 1449663
Topic Name: Chemistry-Section A

Class XII students were asked to prepare one litre of buffer solution of pH 8.26 by their Chemistry teacher. The amount of ammonium chloride to be dissolved by the student in 0.2 M ammonia solution to make one litre of the buffer is
(Given: pK_b (NH_3) = 4.74
Molar mass of NH_3 = 17 g mol^{-1}
Molar mass of NH_4Cl = 53.5 g mol^{-1})

Question:

A 53.5 g

B 72.3 g

C 107.0 g

D 126.0 g

Answer Given By Candidate: **Not Attempted**

Question ID: 1449664
Topic Name: Chemistry-Section A

At 30°C, the half life for the decomposition of AB_2 is 200 s and is independent of the initial concentration of AB_2. The time required for 80% of the AB_2 to decompose is
Given: log 2 = 0.30
log 3 = 0.48

Question:

A 200 s

B 323 s

C 467 s

D 532 s

Answer Given By Candidate: **Not Attempted**

Question ID: 1449665
Topic Name: Chemistry-Section A
Question: Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Finest gold is red in colour, as the size of the particles increases, it appears purple then blue and finally gold.

Reason R: The colour of the colloidal solution depends on the wavelength of light scattered by the dispersed particles.

In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

A Both **A** and **R** are true and **R** is the correct explanation of **A**.

B Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

C **A** is true but **R** is false.

D **A** is false but **R** is true.

Answer Given By Candidate: **A**

Question ID: 1449666
Topic Name: Chemistry-Section A
Question: The metal that has very low melting point and its periodic position is closer to a metalloid is

A Al

B Ga

C Se

D In

Answer Given By Candidate: **B**

Question ID: 1449667
Topic Name: Chemistry-Section A
Question: The metal that is not extracted from its sulfide ore is

A Aluminium

B Iron

C Lead

D Zinc

Answer Given By Candidate: **A**

Question ID: 1449668
Topic Name: Chemistry-Section A

Question: The products obtained from a reaction of hydrogen peroxide and acidified potassium permanganate are

A Mn^{4+}, H_2O only

B Mn^{2+}, H_2O only

C Mn^{4+}, H_2O, O_2 only

D Mn^{2+}, H_2O, O_2 only

Answer Given By Candidate: **D**

Question ID: 1449669
Topic Name: Chemistry-Section A

Question: Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A: LiF is sparingly soluble in water.
Reason R: The ionic radius of Li^+ ion is smallest among its group members, hence has least hydration enthalpy.
In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

A Both **A** and **R** are true and **R** is the correct explanation of **A.**

B Both **A** and **R** are true but **R** is NOT the correct explanation of **A.**

C **A** is true but **R** is false.

D **A** is false but **R** is true.

Answer Given By Candidate: **D**

Question ID: 1449670
Topic Name: Chemistry-Section A

Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Boric acid is a weak acid.

Reason R: Boric acid is not able to release H^+ ion on its own. It receives OH^- ion from water and releases H^+ ion.

In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

Question:

A Both **A** and **R** are correct and **R** is the correct explanation of **A.**

B Both **A** and **R** are correct but **R** is NOT the correct explanation of **A.**

C **A** is correct but **R** is not correct.

D **A** is not correct but **R** is correct.

Answer Given By Candidate: **A**

Question ID: 1449671

Topic Name: Chemistry-Section A

Question: The metal complex that is diamagnetic is (Atomic number: Fe,26; Cu,29)

A $K_3[Cu(CN)_4]$

B $K_2[Cu(CN)_4]$

C $K_3[Fe(CN)_4]$

D $K_4[FeCl_6]$

Answer Given By Candidate: **A**

Question ID: 1449672

Topic Name: Chemistry-Section A

Match **List I** with **List II**.

List I **Pollutant**	**List II** **Source**
A. Microorganisms	I. Strip mining
B. Plant nutrients	II. Domestic sewage
C. Toxic heavy metals	III. Chemical fertilizer
D. Sediment	IV. Chemical factory

Question: Choose the correct answer from the options given below :

A A-II, B-III, C-IV, D-I

B A-II, B-I, C-IV D-III

C A-I, B-IV, C-II, D-III

D A-I, B-IV, C-III, D-II

Answer Given By Candidate:**B**

Question ID:1449673
Topic Name: Chemistry-Section A
Question: The correct decreasing order of priority of functional groups in naming an organic compound as per IUPAC system of nomenclature is

A $-COOH > -CONH_2 > -COCl > -CHO$

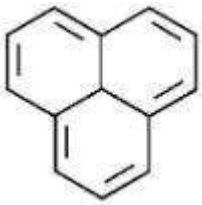
B $-SO_3H > -COCl > -CONH_2 > -CN$

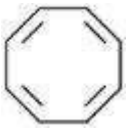
C $-COOR > -COCl > -NH_2 > \rangle C = O$

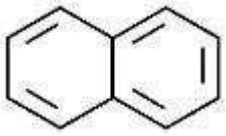
D $-COOH > -COOR > -CONH_2 > -COCl$

Answer Given By Candidate:**B**

Question ID:1449674
Topic Name: Chemistry-Section A
Question: Which of the following is not an example of benzenoid compound?

A 

B 

C 

D 

Answer Given By Candidate:**B**

Question ID:1449675
Topic Name: Chemistry-Section A
Question: Hydrolysis of which compound will give carbolic acid?

A Cumene

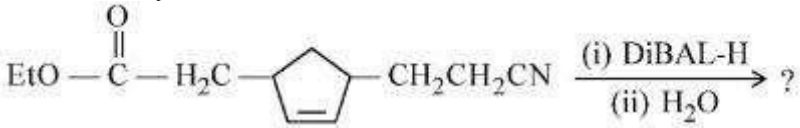
B Benzenediazonium chloride

C Benzal chloride

D Ethylene glycol ketal

Answer Given By Candidate:**B**

Question ID:1449676
Topic Name: Chemistry-Section A



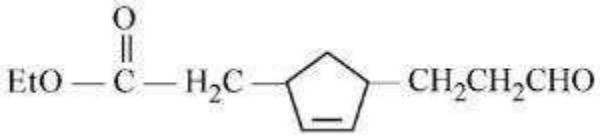
[Et is $-C_2H_5$]

Question: Consider the above reaction and predict the major product.

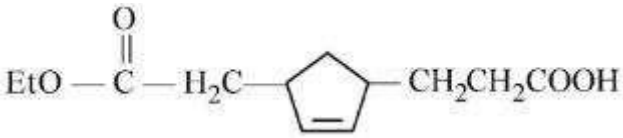
A



B



C



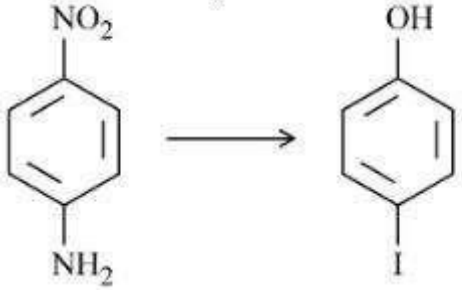
D



Answer Given By Candidate:**D**

Question ID:1449677
Topic Name: Chemistry-Section A
The correct sequential order of the reagents for the given reaction is

Question:



A HNO_2, Fe/H^+, HNO_2, KI, H_2O/H^+

B HNO_2, KI, Fe/H^+, HNO_2, H_2O/warm

C HNO_2, KI, HNO_2, Fe/H^+, H_2O/H^+

D HNO_2, Fe/H^+, KI, HNO_2, H_2O/warm

Answer Given By Candidate: **Not Attempted**

Question ID: 1449678
Topic Name: Chemistry-Section A
Question: Vulcanization of rubber is carried out by heating a mixture of

A isoprene and styrene

B neoprene and sulphur

C isoprene and sulphur

D neoprene and styrene

Answer Given By Candidate: **B**

Question ID: 1449679
Topic Name: Chemistry-Section A
Question: Animal starch is the other name of

A amylose.

B maltose.

C glycogen.

D amylopectin.

Answer Given By Candidate: **Not Attempted**

Question ID: 1449680
Topic Name: Chemistry-Section A
Question: Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A: Phenolphthalein is a pH dependent indicator, remains colourless in acidic solution and gives pink colour in basic medium.
Reason R: Phenolphthalein is a weak acid. It doesn't dissociate in basic medium.
In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

A Both **A** and **R** are true and **R** is the correct explanation of **A.**

B Both **A** and **R** are true but **R** is NOT the correct explanation of **A.**

C **A** is true but **R** is false.

D **A** is false but **R** is true.

Answer Given By Candidate:**C**

Question ID:1449681
Topic Name: Chemistry-Section B

Question: A 10 g mixture of hydrogen and helium is contained in a vessel of capacity 0.0125 m^3 at 6 bar and 27°C. The mass of helium in the mixture is _____ g. (nearest integer)
Given: R = 8.3 $J\ K^{-1}\ mol^{-1}$
(Atomic masses of H and He are 1 u and 4 u, respectively)

Answer Given By Candidate:**7**

Question ID:1449682
Topic Name: Chemistry-Section B

Question: Consider an imaginary ion $^{48}_{22}X^{3-}$. The nucleus contains 'a'% more neutrons than the number of electrons in the ion. The value of 'a' is _____. [nearest integer]

Answer Given By Candidate:**4**

Question ID:1449683
Topic Name: Chemistry-Section B

Question: For the reaction
$H_2F_2(g) \rightarrow H_2(g)+F_2(g)$
$\Delta U = -59.6$ kJ mol^{-1} at 27°C.
The enthalpy change for the above reaction is (–) _____ kJ mol^{-1} [nearest integer]
Given: R = 8.314 $J\ K^{-1} mol^{-1}$.

Answer Given By Candidate:**57**

Question ID:1449684
Topic Name: Chemistry-Section B

Question: The elevation in boiling point for 1 molal solution of non-volatile solute A is 3 K. The depression in freezing point for 2 molal solution of A in the same solvent is 6 K. The ratio of K_b and K_f i.e., K_b/K_f is 1: X. The value of X is [nearest integer]

Answer Given By Candidate:**1**

Question ID:1449685
Topic Name: Chemistry-Section B

Question: 20 mL of 0.02 M hypo solution is used for the titration of 10 mL of copper sulphate solution, in the presence of excess of KI using starch as an indicator. The molarity of Cu^{2+} is found to be _____ $\times 10^{-2}$ M. [nearest integer]

Given : $2\,Cu^{2+} + 4\,I^- \rightarrow Cu_2I_2 + I_2$

$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$

Answer Given By Candidate: **Not Attempted**

Question ID: 1449686
Topic Name: Chemistry-Section B

Question: The number of non-ionisable protons present in the product B obtained from the following reactions is _____.

$C_2H_5OH + PCl_3 \rightarrow C_2H_5Cl + A$

$A + PCl_3 \rightarrow B$

Answer Given By Candidate: **Not Attempted**

Question ID: 1449687
Topic Name: Chemistry-Section B

Question: The spin-only magnetic moment value of the compound with strongest oxidizing ability among MnF_4, MnF_3 and MnF_2 is _____ B.M. [nearest integer]

Answer Given By Candidate: **4**

Question ID: 1449688
Topic Name: Chemistry-Section B

Question: Total number of isomers (including stereoisomers) obtained on monochlorination of methylcyclohexane is _____.

Answer Given By Candidate: **Not Attempted**

Question ID: 1449689
Topic Name: Chemistry-Section B

Question: A 100 mL solution of CH_3CH_2MgBr on treatment with methanol produces 2.24 mL of a gas at STP. The weight of gas produced is _____ mg. [nearest integer]

Answer Given By Candidate: **Not Attempted**

Question ID: 1449690
Topic Name: Chemistry-Section B

Question: How many of the following drugs is/are example(s) of broad spectrum antibiotics ?

Ofloxacin, Penicillin G, Terpineol, Salvarsan.

Answer Given By Candidate: **Not Attempted**